Auditing State-Visitation Bottlenecks in On-Policy Distillation: A Persistent Game-Agent Case Study

Anonymous ACL submission

Abstract

On-policy distillation supervises student-visited states, potentially excluding persistent states where a teacher has an advantage. We audit this failure mode for a 2B agent using 17 tools in a persistent MMORPG. One natural-visitation run changes tool use and tempo but not a 12/30 quest score. A later round adds rollouts from an intermediate player state and is followed by a reported canonical-start score of 15/30. We recover and content-bind nine mechanism-arm runs and reproduce every six-hour score from raw records. Across four historically intended corpus-and-initialization-matched pairs, uniform self-imitation matches or exceeds teacher-graded training in three cells; seeded-corpus arms have higher observed scores than natural-only and poor-corpus controls. These are one-run historical comparisons with incomplete execution provenance, so they locate a visitation/corpus hypothesis rather than estimate an effect. We specify the replicated study needed to test that hypothesis. A separate preregistered 18-cell recovery diagnostic completes, but none of 958 generations creates a recovery opportunity. The contribution is an audited case, a falsifiable protocol, and a failed manipulation check—not an established method effect.

1 Introduction

Post-training a small language model to act over long horizons is not just a token-prediction problem. The model chooses which environment states become training data. If it never reaches a state where its teacher knows what to do, dense teacher feedback is unavailable exactly where it is needed. This creates a state-visitation bottleneck: improving token-level agreement on common states can alter style without transferring the competence that distinguishes the teacher.

We examine that bottleneck in a persistent tool-use setting. A Qwen/Qwen3.5-2B student (Qwen Team, 2026) plays a tile-based MMORPG through a typed action interface. The world persists across conversation rollovers, and benchmark progress requires long chains of gathering, navigation, dialogue, and quest actions. Historical records report a scaffolded Qwen/Qwen3.5-4B teacher run reaching states that the 2B student run does not. We apply reverse-KL on-policy distillation to the student's own emitted tokens. The historical comparison contrasts natural student visitation with a mixture that begins some rollouts from loadable, walkability-checked intermediate database states; the confirmatory protocol additionally freezes a measurable state-selection rule before training.

The present evidence is hypothesis-generating and under-replicated. It contains one complete six-hour run per principal arm; the three agents within a run are prompt variants sharing weights, harness, and launch conditions, not independent experimental units. We therefore separate observations from claims throughout. The observed sequence is: base 12/30, natural-visitation round 12/30, and state-augmented round 15/30. In the last arm, every prompt variant passes the same wall from a canonical-start evaluation world with no intermediate-state initialization. This motivates testing whether training visitation matters, but it does not isolate state seeding from all round-to-round changes or estimate population-level uncertainty.

A recovered July campaign adds controls that materially sharpen, but do not complete, this account. We copy-verify 19,005 files from nine mechanism-arm runs, bind each agent/run directory cryptographically, and replay observations only through its recorded six-hour boundary. Four historical pairs were labelled as retaining the same corpus and initialization while replacing teacher-derived token weights with a constant. The uniform arm matches or exceeds the graded score in three of four cells. A repaired-render graded sensitivity reverses the first pair numerically. A teacher-free arm trained from poorer base-policy rollouts returns to the base score. This pattern points away from a simple "better token weights" explanation and toward which trajectories enter the corpus, but each cell is still one historical training run without a complete checkpoint or execution receipt.

This working draft makes three bounded contributions:

1. We audit a state-visitation failure case and separate observations that survive artifact review from causal claims that do not. We provide a confirmatory protocol with controls capable of falsifying the proposed player-state intervention.

2. We distinguish behavioral imitation from competence in the surviving historical record, and recover four intended graded-versus-uniform pairs that show no consistent score advantage from teacher-derived token weights.

3. We report a preregistered local recovery diagnostic that completes all 18 cells yet produces zero eligible defects. This failed manipulation check prevents an otherwise tempting recovery-effect claim and changes the next experiment.

Separately, one targeted teacher-forcing probe motivates a copy-prior hypothesis at the tool boundary. It does not establish a within-context preference reversal and is not a headline contribution.

The broader lesson is methodological. Model weights, visited states, and runtime affordances are different interventions. A credible agent-training result must preserve and cross them rather than roll them into one headline.

2 Setting

2.1 Persistent tool-use benchmark

The environment is a pinned fork of Kaetram, an open-source tile-based MMORPG (Kaetram contributors, 2026), backed by a game server and MongoDB. Each policy controls three agents with different priority prompts (grinder, completionist, and explorer) through 17 typed tools. Tools cover observation, navigation, combat, gathering, dialogue, quest inspection, inventory use, and crafting. The prompt includes procedural quest guidance, so the benchmark measures execution of a supplied plan rather than autonomous world-model discovery.

The primary development metric, Core-3, counts newly reached stages in three quest chains: Foresting (3 stages), Herbalist's Desperation (3), and Rick's Roll (4). Each agent can gain at most ten stages; a three-agent run therefore scores in $[0, 30]$. Historical notes describe runs as beginning from a freshly initialized player state in a restarted server, but no fail-closed reset receipt or complete shared-world hash survives. Runs continue for six hours while conversation sessions roll over under a fixed context budget and game state persists. The intended confirmatory experimental unit is a full run with an attested reset. Prompt variants are clustered observations within that run.

2.2 Agent and interface contract

The policy acts in an observe–reason–act loop through an OpenAI-compatible model endpoint and a typed tool server. Navigation and selected interactions contain deterministic affordances such as path finding and automatic walking; the model still decides goals, tools, and arguments. The current development record contains an important interface asymmetry: older base-model runs used a native model-visible tool schema while some fine-tuned runs did not receive an identical serialization. We therefore do not use those comparisons as primary evidence. The confirmatory implementation requires normalized full-schema hash parity across training and every evaluation arm.

3 On-Policy Distillation Under Visitation Constraints

Let $x \in \mathcal{X}$ denote persistent player state (including position, inventory, skills, and quest progress) and $h \in \mathcal{H}$ the model-visible interaction history. The joint occupancy state is their joint value $z = (x, h)$; two rollouts with the same database snapshot but different visible histories need not induce the same action distribution. Let μ initialize player states, $q(h \mid x)$ initialize visible histories, and $d^{\pi}_{\mu,q}(x, h)$ denote the joint occupancy induced thereafter by policy $\pi(a \mid h)$ and the environment dynamics, whose transitions depend on the latent player and world state. The model does not observe x directly; it acts only from h, which may contain tool observations derived from x. The canonical serving initializers are μ_0 and q_0. A reverse-KL on-policy objective can be written

$$\mathcal{L}_{\text{OPD}} = \mathbb{E}_{(x,h)\sim d^{\pi_S}_{\mu,q}} \mathbb{E}_{a\sim\pi_S(\cdot|h)} \left[\log \pi_S(a \mid h) - \log \pi_T(a \mid h)\right]. \tag{1}$$

The teacher scores the student's emitted tokens. Training uses a clipped importance-weighted update around the rollout policy, with advantages frozen at data-build time. Full implementation details appear in Appendix A.

Equation 1 makes the bottleneck explicit. If a teacher advantage is concentrated on a set $H \subseteq \mathcal{X} \times \mathcal{H}$ for which $\Pr_{(x,h)\sim d^{\pi_S}_{\mu_0,q_0}}[(x, h) \in H] \approx 0$, token-dense grading supplies almost no information about that advantage. We change the state source, not the loss: some student rollouts begin from intermediate player states, and their records are mixed with natural student rollouts. The student still acts and the same teacher still grades every emitted token. Because changing x can also change the compatible h, the confirmatory design crosses player-state initialization with matched history construction rather than treating a database snapshot as the complete policy state.

The seeded states are typed persistent player states rather than textual demonstrations. Player rows specify quest stage, skill level, inventory, and a walkability-checked position. End-to-end loadability is necessary but not sufficient for legal reachability. Each confirmatory candidate must also carry either a hash-verified witness trajectory from the canonical start that is replayed by a pinned checker, or an invariant certificate executed by a pinned checker that establishes a valid path to the snapshot. Hash continuity or certificate-shaped metadata alone is not sufficient. Evaluation always starts from the canonical initial player state; prospective runs nevertheless record distinct gameplay-RNG seeds. This resembles reset-based curricula in reinforcement learning (Salimans and Chen, 2018; Resnick et al., 2018; Ecoffet et al., 2021). TCOD uses teacher-success prefixes to initialize multi-turn OPD curricula (Wang et al., 2026); Guided-OPD mixes teacher and student turns and anneals guidance to zero (Li et al., 2026a); ReOPD treats prefix selection as a teacher-reliability problem (Liao et al., 2026). These works preempt generic novelty for intermediate starts or state-distribution design. Our narrower object is a direct, prefix-independent player-state initializer. A candidate state need not lie on a successful teacher trajectory and is selected by a frozen rule rather than by progress alone.

Table 1: Frozen targeted-state admission rule. L and U are 95% Wilson bounds recomputed from hash-bound Boolean trials. Every row must pass.

Criterion	$n_{\min}$	Rule
Visit rate	500	$U(v) \leq .05$
Teacher success	100	$L(p_T) \geq .60$
Teacher–student gap	100 each	$L(p_T) - U(p_S) \geq .30$
Recovery	100	$L(p_R) \geq .80$
Validity/reachability	–	pinned check

Let $v(x)$ be estimated natural student visitation, and let $p_T(x, q)$ and $p_S(x, q)$ be repeated teacher and student success rates under a frozen history constructor q. The confirmatory selector admits a player state only if it passes player-state and compatibility invariants, is unfinished and task-relevant, is recoverable, satisfies $v(x) \leq \tau_v$, has certified reachability, and satisfies $p_T(x, q) - p_S(x, q) \geq \tau_\Delta$ together with a minimum teacher success threshold. Counts and denominators, source-run identifiers, complete snapshots, and hash-verified validation artifacts are frozen before outcomes. This rule distinguishes the proposed method from the hand-selected milestone used in the exploratory round.

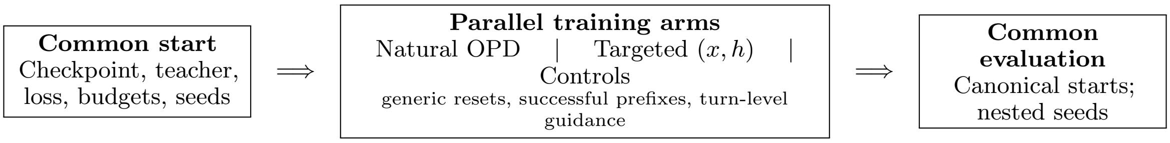

Figure 1: Matched study design. Training arms run in parallel from common initialization; they do not form a sequential curriculum. Model-visible history is crossed explicitly because a database snapshot alone is not the policy state. The fixed-checkpoint weights-by-recovery factorial is a separate diagnostic and cannot estimate this intervention.

4 Evidence Protocol

We grade evidence before interpreting it. For a fixed checkpoint, a fresh-world evaluation run is the independent unit; agents within it share the policy, launch, infrastructure, and analysis pipeline. For a training-method claim, the independent unit is instead a freshly initialized training seed/checkpoint, with evaluation seeds nested within it. Exact scores from single historical runs are descriptive. Counts over agents or sessions diagnose mechanisms but cannot be treated as independent replications of either effect.

The principal historical sequence uses a 2B model lineage, six-hour canonical-start evaluation protocol, benchmark, and nominal harness, but it is not a same-checkpoint comparison. Round 1 trains on natural base-policy rollouts. Round 2 initializes from round 1 and mixes natural round-1 rollouts with 2,326 records collected by the round-1 policy from a loadable, walkability-checked Herbalist milestone state. The round-2 corpus contains 7,849 states, split into 7,024 training and 825 held-out records. Because round 2 also changes the initialization and data batch, round 1 versus round 2 is not a clean state-augmented-versus-natural-only training ablation.

Separately, before observing outcomes we registered a local feasibility pilot using no paid endpoint and crossing Base, round-2, and round-3 public weight snapshots with three paired inference/environment seeds. Each cell uses one completionist prompt identity, recovery off, a canonical tool schema, the same locked tokenizer and patched render contract, and one five-minute canonical-start episode. This pilot tests execution and action throughput only; it is not the six-hour weights-by-recovery factorial and cannot establish policy superiority. We then preregistered a longer 18-cell diagnostic crossing the same three fixed checkpoints with recovery off/on and three paired 30-minute seeds. It remains exploratory and cannot estimate a training procedure.

The *wall* column measures passage of Herbalist stage 1 by the three prompt variants. It is a useful within-run consistency check, not a binomial sample of independent policies. We preserve raw pre-rewrite emissions for confirmatory format analyses; rewritten history is not an authoritative defect counter.

5 Exploratory Results

5.1 Behavior changes without score change in the reported natural-visitation round

Round 1 leaves Core-3 unchanged at 12/30 and every agent reaches the same 4/10 stage profile as base. Nevertheless, turns per session and several aggregate tool shares move toward the 4B teacher: turns/session are 9.1/6.1/6.1 for base/round 1/teacher, observe shares are 26.6/35.5/36.1%, and navigate shares are 14.3/9.4/9.1%. Query-quest share instead overshoots (5.3/18.0/8.7%). These one-run diagnostics are consistent with behavioral imitation on frequently visited states, but they lack matched-state opportunities and uncertainty. We therefore make no quantitative KL–behavior correspondence or token-level causal claim.

5.2 The seeded-training round is followed by canonical-start wall passage

The 4B teacher demonstrates competence beyond the Herbalist stage-1 wall. In the reported base and round-1 runs, zero of three prompt variants crosses that wall; the available record does not provide a defensible denominator for a broader natural-visitation rate. Seeded collection starts the round-1 policy at the accepted quest with the required gathering skill and near

Table 2: Observed development runs and the claims they permit. Scores are descriptive because there is one run per arm. Recovery is a runtime interface affordance.

Policy	Training states	Recovery	Core-3	Wall	Interpretation
Base 2B	–	off	12	0/3	Reference run
OPD round 1	natural	off	12	0/3	Behavior changed; score did not
OPD round 2	natural + seeded	off	15	3/3	Reported recovery-off observation
Round-2 weights	–	on	17	–	Unmatched recovery-on observation
Round-3 weights	broader seeded	on	18	–	Weights plus interface; not pure weights

the target resource. All three seeded collectors pass the wall, creating student-authored actions and teacher scores in this region.

After training on the natural-plus-seeded mixture, round 2 is reported as evaluated from a canonical-start player state in a restarted server, with no intermediate-state initialization and with recovery disabled. Neither a reset receipt nor its historical gameplay-RNG seed was preserved. It scores 15/30, with each prompt variant at 5/10. All three pass Herbalist stage 1, compared with zero of three under both the base and round-1 policies. Logs show earlier quest acceptance, explicit tracking of the skill gate, resource rotation after empty harvests, and less premature return to the quest giver. These are concrete trajectory differences consistent with the intended mechanism.

The valid statement is deliberately narrow: *a seeded-training round was followed by unanimous canonical-start wall passage in one run*. The present data do not show that seeding alone caused the improvement. The required causal experiment trains fresh students from the same checkpoint under identical budgets, seeds, and data construction, varying only whether the reachability-certified state mixture is included (Section 7).

5.3 Recovered controls narrow the mechanism

The July campaign was adaptive development designed after earlier outcomes; none of its arm comparisons is confirmatory. We recovered all 27 agent/run directories from nine later mechanism arms. A source-side checksum list covers 19,005 files; a second manifest hashes each directory; and a deterministic replay includes an observation only when its record timestamp falls within that lane's recorded six-hour budget. The replay reproduces the documented totals for base+recovery

Table 3: Recovered runs labelled as intended pairs in historical notes. Matching variables were not sealed. Each cell is one training run; parentheses give clustered wall passages out of three prompt variants. "Uniform" replaces nonzero teacher-derived advantages with one positive constant. The totals are descriptive, not an effect estimate.

Reported corpus / init	Teacher-graded	Uniform
Natural base / base	12 (0/3)	13 (1/3)
Natural r1 / r1	12 (1/3)	14 (2/3)
Natural+seeded r1 / r1	15 (3/3)	15 (3/3)
Broader seeded / r2	18 (3/3)	17 (3/3)

(12), round-3 without recovery (18), round-2 natural-only (12), round-2 uniform (15), round-1 uniform (13), round-2 natural+uniform (14), teacher-free (12), round-3 uniform (17), and repaired-render round-1 (14). This establishes that the scores are present in the recovered logs. It does not establish the identity of the historical checkpoint, corpus, reset, render contract, or random seeds, which were not sealed at launch.

Table 3 includes every available graded/uniform pair that the historical notes label as sharing corpus and initialization; the pairing rule is post-hoc, and the audit verifies scores rather than treatment identity. Uniform training matches or exceeds the graded arm in three cells, so the family does not show a consistent score advantage from teacher-derived token weights. The reported controls set each nonzero advantage to the corpus mean absolute advantage before evaluation: constants 0.487/0.429/0.431/0.362, with 5,564/7,024/7,024/8,856 records and 174/220/220/277 update steps across the four uniform columns. These settings come from historical manifests and notes, not launch-time receipts.

An important sensitivity is omitted from the intended-pair rows because it changes another variable: a repaired-render graded round-1 run scores 14 (wall 2/3), whereas the old-render uniform round-1 arm scores 13 (1/3). The reversal means that "teacher grades add no value" is not stable to the render repair. The natural-only round-2 arm scores 12 while the corresponding seeded-corpus uniform arm scores 15. A teacher-free arm built instead from poorer base-policy seeded rollouts scores 12 and crosses the wall in zero of three prompt variants. Thus "seeding" is not sufficient: which policy generated the visited trajectories, what those trajectories contain, and data volume remain entangled.

The observations are compatible with a substantial corpus-mediated channel: reset-conditioned student trajectories may carry useful behavior without per-token teacher direction in this lineage. Teacher grading may still alter the behavior policy that later generates useful data, and the uniform pairs do not remove execution-era or artifact-lineage alternatives. Independent training replications under a fully crossed corpus-by-weighting design are required.

5.4 Weights and recovery are historically confounded

The June round-3 run combines broader state seeding with a runtime recovery affordance and scores 18/30. A later recovered run labelled with the same round-3 weights and recovery disabled also replays to 18/30. A separate unmatched run of round-2 weights with recovery reports 17/30. The recovery-off run is important counterevidence to an interface-only explanation, but missing configuration parity, checkpoint/render attestation, reset receipts, and gameplay seeds still prevent a pure-weights attribution. The recovery-off contrast of base 12/30 versus round 2 at 15/30 has the same limitation.

The recovery mechanism detects a dropped malformed call, reconstructs the executable call, rewrites the retained history into canonical form, executes the action, and returns an explicit correction note. Historical notes report 405 rewritten sessions with exactly one recovery marker and no later marker. Recovered rewritten bundles exist, but raw pre-rewrite emissions are absent and the 405 count has not been independently regenerated. It therefore does not show that weights learned the format, that the model self-corrected, or that recovery generalizes to other defects.

5.5 Longer local cells expose progress, but not recovery

The preregistered 30-minute diagnostic completed all 18 Base/R2/R3 $\times$ recovery off/on cells; 1,266 sealed files rehashed without mismatch. Four cells advanced one stage in the registered completionist lane. Across 958 raw generations, 294 contained canonical structured calls and 664 contained none. None was malformed or eligible for recovery, and no recovered call occurred. Thus the recovery switch never applied its intervention: on/off differences are not recovery effects. Three seed blocks over fixed checkpoints also cannot establish superiority. Appendix B reports this run and the preceding five-minute execution pilot. The failed manipulation check rules out spending the full factorial budget unchanged; natural trigger incidence must be shown first, and controlled defect injection is a separate conditional question.

6 A Copy-Prior Hypothesis at the Grading Interface

Round 1 introduces a malformed parameter-key pattern that is absent in the base run. The emitted form places a Boolean value inside the parameter key, so the parser silently drops the intended argument. The training records contain the correct syntax, and historical notes report that the teacher favors the correct delimiter in clean base-policy contexts. After the malformed prefix is already present, the same notes report teacher log probability -0.161 (about 85% probability) for continuing the malformed form, versus -0.253 under the student. This is a positive teacher-over-student distillation advantage for one candidate. The reported correct-form observation uses a different state; without both canonical and malformed candidates scored in each matched context, the record does not establish a preference reversal.

Thus teacher generation and teacher-forced grading are distinct objects, but the historical probe establishes only a candidate failure mode. A teacher that generates canonical syntax from a clean history may still assign a locally useful distillation signal to continuation of a student's defect after that defect enters the context. We call this a *teacher-forcing copy-prior hypothesis*. The current record lacks the paired artifact needed to show that the teacher actively prefers the malformed candidate.

This hypothesis relates to exposure bias (Bengio et al., 2015) and to failures of token-level teacher agreement on degraded prefixes. CCOPD already conditions a teacher on clean, canonical evidence to reduce self-anchored drift in a multi-turn student (Lin et al., 2026); that broad clean-context intervention is not novel here. The narrower tool-syntax hypothesis gives a testable prediction: paired histories differing only in canonical versus malformed prior syntax should change teacher log-odds while holding state and candidate continuations fixed. Section 7 turns that prediction into a multi-state, multi-defect experiment.

7 Confirmatory Study Design

The confirmatory question is simple: do targeted persistent player states improve canonical-start performance beyond other ways of exposing a student to useful failures? Every arm below is prospective. The preparation artifact validates record contracts and common optimization settings, but no verified trained-arm outcome exists in this draft.

What is held fixed. Fresh students begin from the same checkpoint and use the same teacher, optimizer, action-token budget, teacher-scoring budget, tool interface, and training-seed schedule. Each trained checkpoint is evaluated across a frozen set of fresh game worlds that start from the ordinary initial player state. The primary endpoint is total CORE-3 stage gain across the three prompt variants; wall passage is secondary. A trained checkpoint, not an agent or conversation, is the independent unit. Replication is frozen by a prospective two-level power analysis rather than justified by the current launcher example.

What is compared. The primary arms are natural OPD, the targeted-state selector, random valid resets, progress-matched resets, TCOD teacher-success prefixes, Guided-OPD, and TRB. Visitation-only and teacher-advantage-only selectors diagnose which part of the rule matters. Corrected-interface SFT and SCoRe-style first-error prefixes provide supervision baselines (Lyu et al., 2025). The targeted method fails its main claim if the simpler reset controls or the strongest prefix/guidance baseline matches it within the frozen smallest effect of interest.

State or visible history? For each admitted player state, the study compares a minimal canonical observation, its authentic teacher prefix, and a matched reconstructed history. A Backplay condition moves the start backward along a verified trajectory. This separates the restored player state x from information carried in the visible history h. The implementation does not restore the complete shared world—for example every NPC, mob, resource, or concurrent player—so it makes no full-world-state claim.

Secondary questions. Structured supervision is tested separately from state source (Liu et al., 2026). Fixed-checkpoint recovery comparisons preserve raw pre-rewrite emissions and distinguish malformed output, eligibility, harness rewrite, parser acceptance, environment execution, and state change. They cannot show learned self-correction or rescue a failed state-selector result. The copy-prior hypothesis is tested with paired canonical and malformed histories that hold state and candidate continuations fixed; teacher and student log-odds are then reported together. No outcome from these secondary modules is counted as extra evidence for the primary method.

Held-out evaluation. One quest was frozen for evaluation before any held-out outcome was inspected. A no-walkthrough prompt removes supplied game knowledge. Results report arrival at the bottleneck, success conditional on arrival, and downstream completion. No held-out outcome is available yet, and a second environment or model family would still be required for a broad claim.

8 Related Work

GKD establishes on-policy distillation from student-generated sequences (Agarwal et al., 2024). Later analyses show that success depends on both teacher–student compatibility and whether the teacher has useful capability on states the student actually visits (Li et al., 2026b). SAD changes which reasoning and action spans receive supervision (Liu et al., 2026), while OPCD and on-policy expert corrections change the feedback or actor within a student trajectory (Ye et al., 2026; Lauffer et al., 2025). Our proposed intervention instead changes where a student rollout begins while keeping the distillation loss fixed.

The closest work also changes the training-state distribution. TCOD starts from teacher-success prefixes; Guided-OPD mixes teacher and student turns; ReOPD selects reliable prefixes; TRB anneals a teacher-near rollout policy; and SCoRe trains from the first student error (Wang et al., 2026; Li et al., 2026a; Liao et al., 2026; Plyusov et al., 2026; Lyu et al., 2025). These methods rule out a broad novelty claim for intermediate starts. The remaining question is narrower: does a frozen selector over low-visitation, teacher-competent *persistent player states* add value beyond successful prefixes, turn-level guidance, and generic resets? Crossing each player state with its model-visible history tests whether any gain comes from restored world state or simply from information in the prefix. That comparison is prospective in this draft.

Intermediate resets have long been used to expose sparse learning signal in reinforcement learning, including reset-along-demonstration, Backplay, and Go-Explore (Salimans and Chen, 2018; Resnick et al., 2018; Ecoffet et al., 2021). DAgger supplies the complementary lesson that imitation data must cover states induced by the learner (Ross et al., 2011). We apply these ideas to a language-model agent with typed tools, but our historical runs do not establish that the proposed selector outperforms these curriculum families.

The recovered uniform controls also connect to self-imitation and demonstration selection. Self-Imitation Learning reinforces an agent's own previously successful decisions (Oh et al., 2018); offline imitation work likewise shows that resultant states and trajectory quality can be useful criteria for selecting among heterogeneous demonstrations (Yue et al., 2024). Our controls are cruder: they retain all reported records, including failures, and replace token weights with a constant. They therefore motivate a corpus-quality question rather than instantiate either method.

BALROG and Orak evaluate game agents broadly across environments and interfaces (Paglieri et al., 2025; Park et al., 2025); GITM, Voyager, and AgentTrek study planning, skills, curricula, or guided replay in open worlds (Zhu et al., 2024; Wang et al., 2024; Xu et al., 2025). Our evidence is deeper but much narrower: one persistent game, one model family, and supplied quest guidance. The game and MCP-style tool interface are infrastructure, not novelty. Likewise, Fission-GRPO learns from execution errors (Zhang et al., 2026); our recovery switch is a runtime rewrite for a fixed checkpoint and cannot be described as learned self-correction.

9 Conclusion

The audited case motivates a concrete hypothesis: low-occupancy player states may hide useful trajectories from natural OPD. Recovered controls show that uniform self-imitation matches several reported teacher-graded scores when it uses a rich corpus, while poorer seeded data does not; repaired-render graded training also numerically exceeds its nearest uniform sensitivity. This narrows the story from direct score transfer to corpus generation and visitation; it does not establish our selector. Matched resets, strong OPD baselines, replication, and held-out transfer must decide whether it survives.

Limitations

The central limitation is statistical: one run per principal arm cannot support a population claim, and three prompt variants are not independent seeds. The state-seeded round also differs from the preceding round in initialization and data. The environment is a single game with procedural walkthroughs, the teacher and student are from one model family, and stage counts are coarse. Some development analyses depend on logs whose raw inputs are not publicly packaged. The recovered July scores are content-bound and deterministically regenerated, but the launches lack exact checkpoint, training-corpus, reset, render, and RNG attestations. This is therefore an audited case study and protocol, not a completed causal method study. A clean clone must also regenerate every main table before submission.

Direct database construction is environment-specific and may not transfer to systems without restorable state. The historical failure state was selected retrospectively, so selection bias is possible; the prospective selector has no outcome yet. The local studies use only three seed blocks per fixed checkpoint; they were designed for feasibility and manipulation checking, not power, and their outcomes cannot enter the confirmatory estimate. Teacher competence at the target state is observed in a small number of trajectories, not estimated precisely. The syntax probe covers one defect family and should not be generalized to all teacher-forced grading.

Ethical Considerations

The work concerns game agents acting in isolated local worlds. Risks include overstating autonomy, silently baking procedural knowledge into prompts, and mistaking recovered actions for model competence. We mitigate these by labeling scaffolding, logging model-visible schemas and raw emissions, separating runtime recovery from weights, and reporting failed interventions. No human-subject or private-user data are used.

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A Training Details

The settings below are reconstructed from the current implementation and notes written after the historical runs; they are reported settings, not a hash-attested historical training configuration.

The teacher is a scaffolded Qwen/Qwen3.5-4B model and the student begins from Qwen/Qwen3.5-2B (Qwen Team, 2026). Both use the same vocabulary. Student rollouts are reconstructed as the byte-exact serving prompt plus the verbatim emitted continuation. This prefix property avoids re-rendering differences caused by chat-template handling of reasoning delimiters. Teacher and behavior log probabilities are computed on the identical raw string. Every tenth session is held out for development gates.

The update uses PPO-style clipped importance weighting (Schulman et al., 2017) with clip $\epsilon = 0.3$, advantage clamp $[-3, 3]$, and a 1.5 weight on the first third of each session. Training uses one epoch, learning rate 5×10^{-5}, effective batch size 32, and a fresh LoRA initialized over the prior merged checkpoint. Action positions alone receive the language-model head. These choices were development decisions, not independently ablated.

Round 1 contains 5,564 train and 574 held-out records from the base policy's natural rollouts. Round 2 combines 5,523 natural round-1 records and 2,326 seeded records before the 7,024/825 train/held-out split. The seed encodes accepted Herbalist stage 1, Foraging level 5, starter inventory, and a walkability-checked tile near the target resource. Evaluation never loads this state.

B Local Exploratory Diagnostics

Table 4: Preregistered local feasibility pilot. Each snapshot has three paired, five-minute cells. Calls/min uses observed wall time, including the final in-flight generation. No-call counts include thinking-only generations.

Weights	Calls/cell	Calls/min	No-call sum
Base	5, 6, 5	0.968	21
Round 2	1, 5, 6	0.736	22
Round 3	5, 4, 5	0.872	21

Every emitted call had a canonical tool name and JSON-object arguments. The retained stderr logs contain no logged `API error:` line, every final session records duration exhaustion, and all sealed inventory hashes reverify. The per-cell call counts overlap substantially; the pilot was not powered for a superiority test and produced no task progress.

The longer diagnostic produced four cells with one registered single-lane stage of progress, but no malformed or recovery-eligible emission. Recovery was therefore not a realized treatment in any pair. The table is descriptive across three fixed seed blocks; it is not a checkpoint ranking.

C Claim-to-Evidence Boundary

Round-1 behavior. Descriptive, one run: measured behavior changed while score did not.

State-seeded round. Unreplicated: the seeded-training round was followed by canonical-start wall passage in one run.

Recovered controls. Record-level replay reproduces all nine July protocol-boundary scores. Four intended graded/uniform pairs show no consistent graded advantage, but one historical training run per cell and incomplete launch provenance preclude an effect estimate.

Targeted selector. Protocol only: no candidate-state bundle, trained checkpoint, or outcome is currently available.

12 to 15. Reported recovery-off historical contrast; not an effect estimate or clean weights comparison, and exact parity remains unavailable.

18/30. A recovered run labelled round-3 with recovery off also reaches 18/30; checkpoint identity and launch parity remain untested, so this is not a pure-weights effect estimate.

Copy prior. Positive teacher-over-student advantage for one malformed continuation in a targeted historical probe; no matched preference reversal.

Recovery. Historical notes report no repeat marker after correction in rewritten sessions. Recovered rewritten bundles exist, but raw pre-rewrite emissions and an independently regenerated marker count are unavailable.

Local pilot. Nine short paired cells executed through the matched local stack with internally hash-checked artifacts; they provide no model-superiority or quest-performance claim and cannot enter the confirmatory estimate.

Recovery diagnostic. All 18 registered 30-minute cells executed and four advanced one stage, but zero of 958 generations was recovery-eligible. The recovery manipulation did not engage; on/off differences are not recovery effects.

Historical bundles. The recovered external inventories bind the headline R10, OPD, and July mechanism agent-run directories, reproduce the R10 statistics, and regenerate the nine July scores. The raw bundles are not a public reviewer artifact and do not attest the executed reset command, game revision, checkpoint identity, or OPD training inputs.

Cross-task transfer. Unsupported; no claim.

D Reproducibility Contract

Before confirmatory runs, each immutable run bundle must contain: environment and game revisions; container or dependency locks; model,

Table 5: Preregistered 30-minute single-lane local weights × recovery exploratory factorial. The structured-call rate pools generations within each arm; progress is the number of seed blocks (out of three) that advanced one registered completionist-lane stage. Exact per-cell values are released in `cells.csv`. No arm produced a recovery-eligible emission. Descriptive only.

Weights	Recovery	Mean calls/min	Pooled structured rate	Eligible emissions	Progress cells
Base	off	0.718	38.5%	0	2/3
Base	on	0.626	35.0%	0	1/3
Round 2	off	0.415	24.8%	0	0/3
Round 2	on	0.457	26.4%	0	0/3
Round 3	off	0.376	23.1%	0	1/3
Round 3	on	0.635	34.7%	0	0/3

tokenizer, adapter, and dataset revisions; complete model-visible tool schemas; rendered-prompt hashes; sampling parameters; environment and inference seeds; raw emissions before any rewrite; state-transition logs; analysis outputs; and checksums. Dataset records must point to source runs and transformations. A clean-clone command must verify hashes and regenerate every table and figure without network-dependent mutable inputs.

The current artifact does not yet meet this bar. One internal recovery hashes 60 selected directories (18,812 files) and regenerates the R10 statistics. A second binds 27 July agent/run directories (19,005 source files) and deterministically regenerates nine six-hour mechanism-arm scores. The raw bundles are not in the reviewer artifact, and neither recovery can attest the historical reset command, game revision, checkpoint identity, OPD training inputs, or render contract. One historical lane also reset a different database from its game servers, so its results are quarantined. The headline R10 and OPD values came from a separate, database-aligned orchestrator and all 36 recovered first states match the canonical level-1 start, but their exact reset command and server revision remain unattested.

A fresh clone now installs and passes the prospective CPU unit suite. Separately, nine five-minute and eighteen 30-minute matched public-weight cells bind the game revision, tokenizer, render contract, seeds, initial state, and sealed inventories. Anonymous CSVs and a public table derivative are checked in, while the complete report, receipt, and raw bundles remain local. This establishes internal local execution and table generation, not independent reproduction, training, or confirmatory evidence.

Future local launches fail closed on the effective database, mode, configuration, and complete model snapshot. Those protections do not retroactively promote older v1 bundles or the completed exploratory factorial. The matched-training contract freezes the arm–seed cells, shared budgets and interfaces, held-out registration, snapshot, tokenizer, and leakage checks. Corrected-interface SFT and first-stage SCoRe-style preparation exist; verified training bundles, SCoRe's reward stage, live Guided-OPD collection, and accelerator results do not. Release packaging, license review, training, and clean-clone regeneration remain submission blockers.

E Planned Statistical Analysis

No session- or agent-level pseudo-replication will be used. For the causal training comparison, a freshly trained checkpoint is the independent unit and fresh-world evaluation seeds are nested repeated measurements. The primary method contrast must aggregate the frozen nested evaluations within each training seed before uncertainty across training seeds is estimated. The five seeds in the current launcher example are not a power justification.

For the fixed-checkpoint weights-by-recovery study, the independent unit is the paired fresh-world replicate cluster; inference is conditional on the three registered artifacts and cannot estimate training-seed variability. Its seven contrasts, 20 clusters, familywise error rule, and assumption-based power record are frozen prospectively. An executable contract fixes the estimator, intervals, effect scale, zero-variance behavior, and no-rerun rule before outcome access. Completion hash-binds the prelaunch ledger, and analysis requires the same clean registered commit while emitting only portable artifact paths. For future trained-model comparisons, the smallest effect and both replication levels must be set from independent pilot variance or a conservative frozen grid. If variance is unknown, use blinded re-estimation with a fixed maximum size. Primary tests are two-sided unless direction and stopping are preregistered. Report every run and exclusion, raw and standardized primary effects, and design-matched uncertainty. Any inferential secondary family needs its own prespecified multiplicity rule; otherwise it remains descriptive. Small wall-passage counts remain descriptive.